Homework Solutions: Lagrangians, Hessians, and Bordered Hessians

GSE 510

Practice Problems: Work together in class

1. Consider the unconstrained function

$$f(x, y) = 20 + 6x + 8y - x^2 - 2y^2.$$

(a) The first partial derivatives are

$$f_x = 6 - 2x$$

and

$$f_y = 8 - 4y.$$

(b) The critical point solves

$$6 - 2x = 0, \qquad 8 - 4y = 0.$$

Thus,

$$x = 3, \qquad y = 2.$$

The critical point is

$$(3, 2).$$

(c) The second partial derivatives are

$$f_{xx} = -2, \qquad f_{yy} = -4, \qquad f_{xy} = 0.$$

Therefore, the Hessian matrix is

$$H = \begin{bmatrix} -2 & 0 \\ 0 & -4 \end{bmatrix}.$$

(d) The determinant is

$$|H| = (-2)(-4) - 0(0) = 8.$$

Since

$$|H| > 0$$

and

$$f_{xx} = -2 < 0,$$

the critical point is a local maximum.

(e) The value at the critical point is

$$f(3, 2) = 20 + 6(3) + 8(2) - 3^2 - 2(2)^2.$$

Therefore,

$$f(3, 2) = 20 + 18 + 16 - 9 - 8 = 37.$$

2. Consider the constrained maximization problem

$$\max_{x,y} xy$$

subject to

$$x + y = 10.$$

(a) Write the Lagrangian as

$$\mathcal{L}(x, y, \lambda) = xy + \lambda(10 - x - y).$$

(b) The first-order conditions are

$$\mathcal{L}_x = y - \lambda = 0,$$
$$\mathcal{L}_y = x - \lambda = 0,$$

and

$$\mathcal{L}_\lambda = 10 - x - y = 0.$$

(c) From the first two conditions,

$$y = \lambda$$

and

$$x = \lambda.$$

Hence,

$$x = y.$$

Using the constraint,

$$x + y = 10,$$

we get

$$2x = 10,$$

so

$$x^* = 5, \qquad y^* = 5.$$

Since $x = \lambda$, we also get

$$\lambda^* = 5.$$

(d) Geometrically, we are maximizing the product xy along the line

$$x + y = 10.$$

The product is maximized when the two variables are balanced equally:

$$x = y = 5.$$

(e) The maximum value is

$$x^* y^* = 5(5) = 25.$$

3. Consider the consumer problem

$$\max_{x,y} x^{1/2} y^{1/2}$$

subject to

$$2x + y = 12.$$

(a) The Lagrangian is

$$\mathcal{L}(x, y, \lambda) = x^{1/2} y^{1/2} + \lambda(12 - 2x - y).$$

(b) The first-order conditions are

$$\mathcal{L}_x = \frac{1}{2} x^{-1/2} y^{1/2} - 2\lambda = 0,$$
$$\mathcal{L}_y = \frac{1}{2} x^{1/2} y^{-1/2} - \lambda = 0,$$

and

$$\mathcal{L}_\lambda = 12 - 2x - y = 0.$$

(c) From the first two first-order conditions,

$$\frac{1}{2}x^{-1/2}y^{1/2} = 2\lambda$$

and

$$\frac{1}{2}x^{1/2}y^{-1/2} = \lambda.$$

Dividing the marginal utilities gives the tangency condition:

$$\frac{MU_x}{MU_y} = \frac{2}{1} = 2.$$

Now

$$\frac{MU_x}{MU_y} = \frac{\frac{1}{2}x^{-1/2}y^{1/2}}{\frac{1}{2}x^{1/2}y^{-1/2}} = \frac{y}{x}.$$

Therefore,

$$\frac{y}{x} = 2,$$

so

$$y = 2x.$$

(d) Use the budget constraint:

$$2x + y = 12.$$

Substitute $y = 2x$:

$$2x + 2x = 12.$$

Hence,

$$4x = 12,$$

so

$$x^* = 3.$$

Then

$$y^* = 2(3) = 6.$$

The optimal bundle is

$$(x^*, y^*) = (3, 6).$$

(e) Using

$$\lambda = \frac{1}{2}x^{1/2}y^{-1/2},$$

we get

$$\lambda^* = \frac{1}{2}\sqrt{\frac{x^*}{y^*}} = \frac{1}{2}\sqrt{\frac{3}{6}} = \frac{1}{2}\sqrt{\frac{1}{2}} = \frac{1}{2\sqrt{2}}.$$

4. Consider the problem

$$\max_{x,y} f(x, y) = xy$$

subject to

$$x + y = 10.$$

The Lagrangian is

$$\mathcal{L}(x, y, \lambda) = xy + \lambda(10 - x - y).$$

(a) The Hessian matrix of the objective function is

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

(b) The constraint can be written as

$$g(x, y) = x + y = 10.$$

Thus,

$$g_x = 1, \qquad g_y = 1.$$

The second derivatives of the Lagrangian with respect to x and y are

$$\mathcal{L}_{xx} = 0, \qquad \mathcal{L}_{yy} = 0, \qquad \mathcal{L}_{xy} = 1.$$

Therefore, the bordered Hessian is

$$B = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

(c) The critical point from Question 2 is

$$(x^*, y^*) = (5, 5).$$

Since the bordered Hessian is constant, evaluating it at the critical point gives the same matrix:

$$B(5, 5) = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

(d) The determinant is

$$|B| = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 2.$$

Using the sign convention discussed in class for this setup, a positive bordered Hessian determinant indicates a constrained maximum.

(e) This matches the intuition from Question 2. Along the line $x + y = 10$, the product xy is largest when x and y are equal, so the constrained maximum occurs at

$$(5, 5).$$

5. Consider the constrained minimization problem

$$\min_{x,y} x^2 + y^2$$

subject to

$$x + y = 6.$$

(a) The Lagrangian is

$$\mathcal{L}(x, y, \lambda) = x^2 + y^2 + \lambda(6 - x - y).$$

(b) The first-order conditions are

$$\mathcal{L}_x = 2x - \lambda = 0,$$
$$\mathcal{L}_y = 2y - \lambda = 0,$$

and

$$\mathcal{L}_\lambda = 6 - x - y = 0.$$

(c) From the first two conditions,

$$2x = \lambda$$

and

$$2y = \lambda.$$

Therefore,

$$x = y.$$

Using the constraint,

$$x + y = 6,$$

we get

$$2x = 6,$$

so

$$x^* = 3, \qquad y^* = 3.$$

(d) The constraint is

$$g(x, y) = x + y = 6,$$

so

$$g_x = 1, \qquad g_y = 1.$$

Also,

$$\mathcal{L}_{xx} = 2, \qquad \mathcal{L}_{yy} = 2, \qquad \mathcal{L}_{xy} = 0.$$

Thus, the bordered Hessian is

$$B = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}.$$

(e) The determinant is

$$|B| = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{vmatrix} = -4.$$

Using the sign convention discussed in class, this indicates a constrained minimum. This also makes intuitive sense: among all points on the line $x + y = 6$, the point $(3, 3)$ is closest to the origin and therefore minimizes $x^2 + y^2$.

Homework Problems: Submit individually

6. Consider the unconstrained function

$$f(x,y) = 4x^2 + 2y^2 - 4xy - 8x - 4y.$$

(a) The first partial derivatives are

$$f_x = 8x - 4y - 8$$

and

$$f_y = 4y - 4x - 4.$$

(b) The first-order conditions are

$$8x - 4y - 8 = 0$$

and

$$4y - 4x - 4 = 0.$$

Divide both equations by 4:

$$2x - y - 2 = 0$$

and

$$y - x - 1 = 0.$$

From the second equation,

$$y = x + 1.$$

Substitute into the first equation:

$$2x - (x+1) - 2 = 0.$$

Thus,

$$x - 3 = 0,$$

so

$$x = 3.$$

Then

$$y = 3 + 1 = 4.$$

The critical point is

$$(3, 4).$$

(c) The second partial derivatives are

$$f_{xx} = 8, \qquad f_{yy} = 4, \qquad f_{xy} = -4.$$

Therefore,

$$H = \begin{bmatrix} 8 & -4 \\ -4 & 4 \end{bmatrix}.$$

(d) The determinant of the Hessian is

$$|H| = 8(4) - (-4)(-4) = 32 - 16 = 16.$$

(e) Since

$$|H| > 0$$

and

$$f_{xx} = 8 > 0,$$

the critical point is a local minimum.

(f) The value at the critical point is

$$f(3,4) = 4(3)^2 + 2(4)^2 - 4(3)(4) - 8(3) - 4(4).$$

Therefore,

$$f(3,4) = 36 + 32 - 48 - 24 - 16 = -20.$$

7. Consider the consumer problem

$$\max_{x,y} x^{1/3}y^{2/3}$$

subject to

$$x + 2y = 18.$$

(a) The Lagrangian is

$$\mathcal{L}(x, y, \lambda) = x^{1/3}y^{2/3} + \lambda(18 - x - 2y).$$

(b) The first-order conditions are

$$\mathcal{L}_x = \frac{1}{3}x^{-2/3}y^{2/3} - \lambda = 0,$$

$$\mathcal{L}_y = \frac{2}{3}x^{1/3}y^{-1/3} - 2\lambda = 0,$$

and

$$\mathcal{L}_\lambda = 18 - x - 2y = 0.$$

(c) From the first-order conditions,

$$MU_x = \lambda$$

and

$$MU_y = 2\lambda.$$

Therefore,

$$\frac{MU_x}{MU_y} = \frac{1}{2}.$$

Now

$$\frac{MU_x}{MU_y} = \frac{\frac{1}{3}x^{-2/3}y^{2/3}}{\frac{2}{3}x^{1/3}y^{-1/3}} = \frac{1}{2}\frac{y}{x}.$$

Hence,

$$\frac{1}{2}\frac{y}{x} = \frac{1}{2}.$$

Therefore,

$$y = x.$$

(d) Use the constraint:

$$x + 2y = 18.$$

Since $y = x$,

$$x + 2x = 18.$$

Thus,

$$3x = 18,$$

so

$$x^* = 6.$$

Hence,

$$y^* = 6.$$

The optimal bundle is

$$(x^*, y^*) = (6, 6).$$

(e) The maximum utility is

$$u(6, 6) = 6^{1/3}6^{2/3} = 6.$$

(f) The Lagrange multiplier measures the marginal value of relaxing the constraint. Here, it tells us approximately how much maximum utility would increase if the consumer's available budget, or right-hand side of the constraint, increased by one unit.

8. Consider the firm problem

$$\max_{L,K} 40L + 60K - L^2 - 2K^2 - LK$$

subject to

$$L + K = 20.$$

(a) The Lagrangian is

$$\mathcal{L}(L, K, \lambda) = 40L + 60K - L^2 - 2K^2 - LK + \lambda(20 - L - K).$$

(b) The first-order conditions are

$$\mathcal{L}_L = 40 - 2L - K - \lambda = 0,$$

$$\mathcal{L}_K = 60 - 4K - L - \lambda = 0,$$

and

$$\mathcal{L}_\lambda = 20 - L - K = 0.$$

(c) From the first two equations,

$$40 - 2L - K = 60 - 4K - L.$$

Simplifying,

$$-20 - L + 3K = 0.$$

Therefore,

$$3K - L = 20.$$

The constraint gives

$$L + K = 20.$$

From the constraint,

$$L = 20 - K.$$

Substitute into $3K - L = 20$:

$$3K - (20 - K) = 20.$$

Thus,

$$4K - 20 = 20,$$

so

$$4K = 40, \qquad K^* = 10.$$

Then

$$L^* = 20 - 10 = 10.$$

To find λ^*, use

$$40 - 2L - K - \lambda = 0.$$

Substituting $L = 10$, $K = 10$:

$$40 - 20 - 10 - \lambda = 0.$$

Therefore,

$$10 - \lambda = 0,$$

so

$$\lambda^* = 10.$$

(d) The Hessian matrix of the objective function is

$$H = \begin{bmatrix} -2 & -1 \\ -1 & -4 \end{bmatrix}.$$

(e) The constraint is

$$g(L,K) = L + K = 20,$$

so

$$g_L = 1, \qquad g_K = 1.$$

The second derivatives of the Lagrangian with respect to L and K are

$$\mathcal{L}_{LL} = -2, \qquad \mathcal{L}_{KK} = -4, \qquad \mathcal{L}_{LK} = -1.$$

Therefore, the bordered Hessian is

$$B = \begin{bmatrix} 0 & 1 & 1 \\ 1 & -2 & -1 \\ 1 & -1 & -4 \end{bmatrix}.$$

(f) The determinant is

$$|B| = \begin{vmatrix} 0 & 1 & 1 \\ 1 & -2 & -1 \\ 1 & -1 & -4 \end{vmatrix} = 4.$$

Using the sign convention discussed in class, this indicates a constrained maximum.

(g) The constraint

$$L + K = 20$$

means the firm has a fixed total amount of input resources to allocate between labor and capital. Choosing more labor requires choosing less capital, and vice versa.

9. **A case where the Lagrangian answer is not the actual economic solution.**

A consumer has utility

$$u(x,y) = x + 2y$$

and budget constraint

$$x + y = 10,$$

with

$$x \geq 0, \qquad y \geq 0.$$

(a) Ignoring the nonnegativity constraints, the Lagrangian is

$$\mathcal{L}(x,y,\lambda) = x + 2y + \lambda(10 - x - y).$$

(b) The first-order conditions are

$$\mathcal{L}_x = 1 - \lambda = 0,$$
$$\mathcal{L}_y = 2 - \lambda = 0,$$

and

$$\mathcal{L}_\lambda = 10 - x - y = 0.$$

(c) The first two conditions imply

$$\lambda = 1$$

and

$$\lambda = 2.$$

These cannot both be true. Therefore, there is no interior solution satisfying the usual Lagrangian first-order conditions. The reason is that the consumer always values good y more than good x, so the optimum occurs at a boundary or corner rather than at an interior tangency.

(d) At $(10,0)$,

$$u(10,0) = 10 + 2(0) = 10.$$

At $(0,10)$,

$$u(0,10) = 0 + 2(10) = 20.$$

(e) The actual utility-maximizing bundle is

$$(x^*, y^*) = (0, 10).$$

(f) This is a corner solution because the consumer spends everything on good y and consumes zero of good x. The optimal bundle lies at the endpoint of the budget line, not at an interior tangency.

(g) The lesson is that Lagrangians are very useful for finding interior candidates, but they are not the whole story. In economic problems with nonnegativity constraints, we must also check boundary and corner solutions.

10. **Slightly harder problem.** Consider the constrained optimization problem

$$\max_{x,y} f(x, y) = x^2 y$$

subject to

$$x + y = 12,$$

with

$$x \geq 0, \qquad y \geq 0.$$

(a) The Lagrangian is

$$\mathcal{L}(x, y, \lambda) = x^2 y + \lambda(12 - x - y).$$

(b) The first-order conditions for an interior solution are

$$\mathcal{L}_x = 2xy - \lambda = 0,$$

$$\mathcal{L}_y = x^2 - \lambda = 0,$$

and

$$\mathcal{L}_\lambda = 12 - x - y = 0.$$

(c) From the first two conditions,

$$2xy = \lambda$$

and

$$x^2 = \lambda.$$

Therefore,

$$2xy = x^2.$$

Since we are looking for an interior solution, $x > 0$, so divide by x:

$$2y = x.$$

Thus,

$$x = 2y.$$

Substitute into the constraint:

$$x + y = 12.$$

Then

$$2y + y = 12,$$

so

$$3y = 12.$$

Hence,

$$y^* = 4, \qquad x^* = 8.$$

The interior critical point is

$$(8, 4).$$

(d) The objective value at the interior critical point is

$$f(8,4) = 8^2(4) = 64(4) = 256.$$

(e) At the endpoint $(12,0)$,

$$f(12,0) = 12^2(0) = 0.$$

At the endpoint $(0,12)$,

$$f(0,12) = 0^2(12) = 0.$$

(f) Yes, the interior critical point is the global maximum on the feasible set. The endpoint values are both 0, while the interior point gives value 256. Since the feasible set is the closed line segment from $(12,0)$ to $(0,12)$, checking the interior critical point and endpoints is enough here.

(g) The constraint is

$$g(x,y) = x + y = 12,$$

so

$$g_x = 1, \qquad g_y = 1.$$

The second derivatives of the Lagrangian with respect to x and y are

$$\mathcal{L}_{xx} = 2y, \qquad \mathcal{L}_{yy} = 0, \qquad \mathcal{L}_{xy} = 2x.$$

At $(8,4)$,

$$\mathcal{L}_{xx} = 8, \qquad \mathcal{L}_{yy} = 0, \qquad \mathcal{L}_{xy} = 16.$$

Therefore, the bordered Hessian at $(8,4)$ is

$$B(8,4) = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 8 & 16 \\ 1 & 16 & 0 \end{bmatrix}.$$

Its determinant is

$$|B(8,4)| = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 8 & 16 \\ 1 & 16 & 0 \end{vmatrix} = 24.$$

Using the sign convention discussed in class, this is consistent with a constrained maximum.

(h) Checking the boundary matters because the Lagrangian first-order conditions only identify interior candidates. If the true optimum lies at a corner or endpoint, the usual interior first-order conditions may fail or may miss the actual economic solution.